

No.

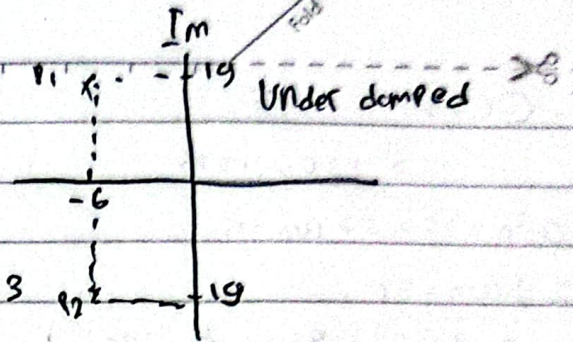
Date

Pole, Respon, dan kestabilannya

$$1. G(s) = \frac{400}{s^2 + 12s + 400}$$

$$\omega_n^2 = 400 \Rightarrow \omega_n = 20$$

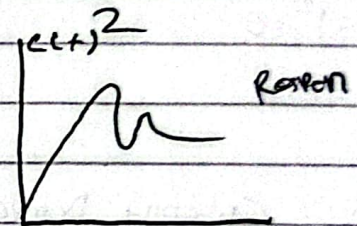
$$2\zeta\omega_n = 12 \Rightarrow 2\zeta(20) = 12 \Rightarrow \zeta = \frac{12}{40} = 0.3$$



Men cari Pole

$$s = \frac{-12 \pm \sqrt{12^2 - 4(1)(400)}}{2} = \frac{-12 \pm \sqrt{144 - 1600}}{2} = \frac{-12 \pm \sqrt{-1456}}{2}$$

$$s_{1,2} = -6 \pm j19$$



Respon dan kestabilan

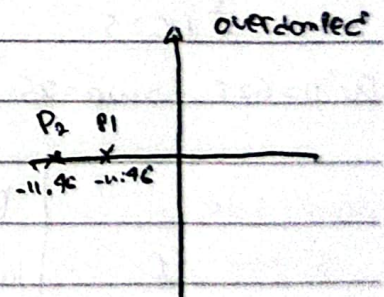
Respon: Karena $0 < \zeta < 1$ (0.3), responnya adalah Underdamped

Kestabilan: Stabil (karena bagian riilnya negatif, yaitu -6)

$$b. G(s) = \frac{900}{s^2 + 90s + 900}$$

$$\omega_n^2 = 900 \Rightarrow \omega_n = 30$$

$$2\zeta\omega_n = 90 \Rightarrow 2\zeta(30) = 90 \Rightarrow \zeta = \frac{90}{60} = 1.5$$



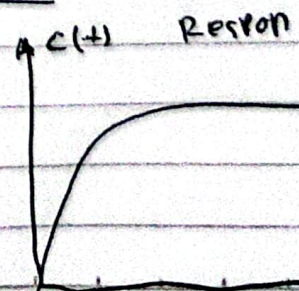
Men cari Pole

$$s = \frac{-90 \pm \sqrt{90^2 - 4(1)(900)}}{2} = \frac{-90 \pm \sqrt{8100 - 3600}}{2} = \frac{-90 \pm \sqrt{4500}}{2}$$

$$s_1 = \frac{-90 + 67.08}{2}$$

$$s_2 = \frac{-90 - 67.08}{2} = -78.54$$

$$s_1 = \frac{-90 + 67.08}{2} = -11.46$$

Respon: overdamped karena $\zeta > 1$ (1.5).

Kestabilan: Stabil (karena Pole, bernilai negatif)

C. $G(s) = \frac{225}{s^2 + 30s + 225}$

$$\omega_n^2 = 225 \Rightarrow \omega_n = 15$$

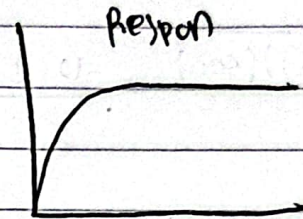
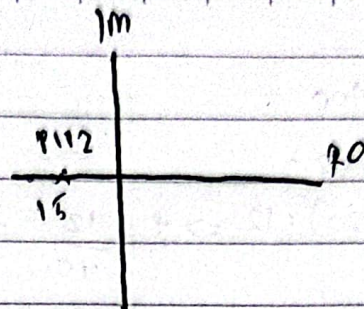
$$2 \zeta \omega_n = 30$$

$$\Rightarrow 2 \zeta (15) = 30 \Rightarrow \zeta = \frac{30}{30} = 1$$

Mencari Pole

$$(s + 15)^2 = 0$$

$$s_{1,2} = -15$$

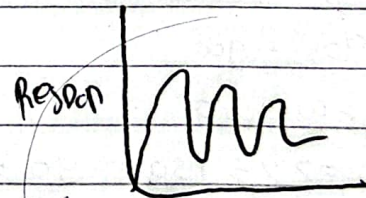
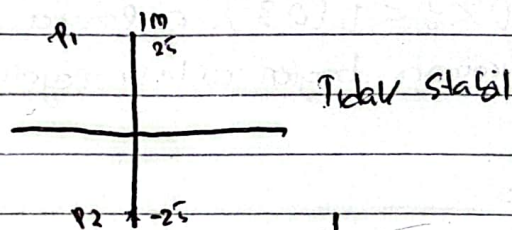


Respon : Critically Damped Karena $\zeta = 1$

Kestabilan : Stabil

d. $G(s) = \frac{625}{s^2 + 625}$

$$\omega_n^2 = 625 \Rightarrow \omega_n = 25$$



Mencari pole

$$s^2 + 625 = 0$$

$$s^2 = -625$$

$$s_{1,2} = \pm j 25$$

Respon = Undamped

Kestabilan = Stabil Marginal /